

1. Probability

Probability measures the chance that an event occurs.

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

where:

- $0 \leq P(E) \leq 1$
 - $P(E) = 0$: Impossible event
 - $P(E) = 1$: Certain event
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2. Sample Space (S)

The set of all possible outcomes.

Example:

Rolling a die

$$S = \{1,2,3,4,5,6\}$$

3. Event (E)

A subset of the sample space.

Example:

Getting an even number

$$E = \{2,4,6\}$$

4. Types of Events

- Simple event: One outcome
 - Compound event: More than one outcome
 - Impossible event: Probability = 0
 - Certain event: Probability = 1
 - Complementary event: Event does not occur
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5. Complement Rule

$$P(A') = 1 - P(A)$$

Example:

If $P(A) = 0.7$,

$$P(A') = 1 - 0.7 = 0.3$$

6. Addition Rule (Union)

For any two events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.55 + 0.45 - 0.20 = 0.80$$

$$P(A)$$

$$P(B)$$

$$P(A \cap B)$$

AB0.20

If events are mutually exclusive,

$$P(A \cup B) = P(A) + P(B)$$

7. Intersection of Events

$$P(A \cap B)$$

$$P(A \cap B) = 0.60 \cdot 0.35 = 0.21$$

$$P(A)$$

$$P(B | A)$$

$$P(A) = 0.60 \quad P(B | A) = 0.35 \quad 0.21$$

For independent events:

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cap B) = P(A)P(B) = (0.60)(0.45) = 0.27$$

$$P(A)$$

$$P(B)$$

$$P(A) = 0.60 \quad P(B) = 0.45 \quad A \cap B = 0.27$$

8. Conditional Probability

Probability of event A occurring given that B has occurred.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.30}{0.65} = 0.46$$

$$P(B)$$

$$P(A \cap B)$$

$P(B)=0.65$ $P(A \cap B)=0.30$ $P(A|B) \approx 0.46$ $A \cap B$ is the part of B where A also happens

1. Tossing One Coin

Question:

A coin is tossed once. What is the probability of getting a Head?

Solution:

Sample Space = {H, T}

Total outcomes = 2

Favourable outcomes = {H} = 1

$$P(\text{Head}) = \frac{1}{2}$$

Answer: $1/2 = 0.5$

2. Rolling One Die

Question:

A die is rolled once. Find the probability of getting 4.

Solution:

Sample Space = {1,2,3,4,5,6}

Total outcomes = 6

Favourable outcomes = {4} = 1

$$P(4) = \frac{1}{6}$$

Answer: $1/6$

3. Even Number on a Die

Question:

Find the probability of getting an even number.

Solution:

Even numbers = {2,4,6}

Favourable outcomes = 3

Total outcomes = 6

$$P(\text{Even}) = \frac{3}{6} = \frac{1}{2}$$

Answer: 1/2

4. Odd Number on a Die

Question:

Find the probability of getting an odd number.

Solution:

Odd numbers = {1,3,5}

Favourable outcomes = 3

Total outcomes = 6

$$P(\text{Odd}) = \frac{3}{6} = \frac{1}{2}$$

Answer: 1/2

5. Drawing a Card

Question:

One card is drawn from a deck of 52 cards. Find the probability of drawing an Ace.

Solution:

Total cards = 52

Number of Aces = 4

$$P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$$

Answer: 1/13

6. Selecting a Student

Question:

A class has 18 boys and 12 girls. One student is selected randomly. Find the probability of selecting a girl.

Solution:

Total students = $18 + 12 = 30$

Girls = 12

$$P(\text{Girl}) = \frac{12}{30} = \frac{2}{5}$$

Answer: 2/5

7. Picking a Marble

Question:

A bag contains 5 red, 3 blue, and 2 green marbles. Find the probability of selecting a blue marble.

Solution:

Total marbles = $5 + 3 + 2 = 10$

Blue marbles = 3

$$P(\text{Blue}) = \frac{3}{10}$$

Answer: 3/10

8. Two Coins Tossed

Question:

Two coins are tossed. Find the probability of getting exactly one Head.

Solution:

Sample Space = {HH, HT, TH, TT}

Total outcomes = 4

Exactly one Head = {HT, TH}

Favourable outcomes = 2

$$P = \frac{2}{4} = \frac{1}{2}$$

Answer: 1/2

